

KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY

Department of Energy Science and Engineering

B. Sc. Engineering 4th Year 2nd Term Examination, 2020

ESE 4003

(Hydrogen and Fuel Cells)

Time: 1 Hour 30 Minutes.

Full Marks: 120

N.B. i) Answer any Two questions from each section.

ii) Figures in the right margin indicate full marks.

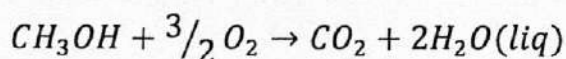
iii) Assume reasonable data if any missing.

SECTION – A

- 1(a). What are the advantages and disadvantages of using hydrogen as fuel? 06
- 1(b). What is water electrolysis? Explain the method of water electrolysis for hydrogen production. 14
- 1(c). Briefly explain the Steam Methane Reforming (SMR). 10
- 2(a). Define biohydrogen. What are the factors affecting nitrogenase-mediated photo fermentation? 06
- 2(b). What are the methods of hydrogen purification? Describe the working principle of membrane separation process. 16
- 2(c). What are the advantages and disadvantages of TiO_2 as photocatalyst for water splitting process? 08
- 3(a). What are the various technologies for hydrogen storage? Write short note on hydride hydrogen storage. 10
- 3(b). Why is it necessary to re-design fuel delivery system of IC engine for hydrogen fuel? Briefly discuss any one method of hydrogen fuel delivery system. 12
- 3(c). Define hydrogen safety. Write down the environmental impacts of hydrogen. 08

SECTION – B

- 4(a). With a neat sketch, explain the working principle of a hydrogen-oxygen fuel cell. 10
- 4(b). Tabulate the differences between battery and fuel cell. Describe the fabrication methods for GDEs. 12
- 4(c). A direct methanol fuel cell uses methanol as fuel. Where, temperature is 30°C and pressure is 100 kPa. Calculate: (i) Δh_{rxn}^0 (ii) Δs_{rxn}^0 and (iii) Δg_{rxn}^0 . 08



Chemic Species	Δh_f^0 (kJ/mol)	Δs_f^0 (J/mol.K)
CH ₃ OH	-200.95	239.83
O ₂	0	205.14
CO ₂	-393.51	213.8
H ₂ O (liq)	-285.83	69.95

- 5(a). What are the benefits of Alkaline Fuel cell (AFC) over acidic fuel cell? 08
- 5(b). Explain the design of a combined heat and power (CHP) fuel cell system with necessary diagram. 12
- 5(c). Write short notes on TPB and MCO. What are the alternative fuels used in fuel cells? 10
- 6(a). What is Gibbs free energy? Derive an expression for maximum possible thermodynamic efficiency of a fuel cell. 10
- 6(b). What is fuel cell subsystem? Why the gross fuel cell stack electrical efficiency differs from the net fuel cell subsystem electrical efficiency? Explain. 12
- 6(c). Assuming 100% fuel utilization, how much current can a fuel cell produce if provisioned with 6 sccm gas at STP? (sccm = standard cubic centimeter per minute, $R = 0.082 \text{ L.atm.K}^{-1}.\text{mol}^{-1}$). Also assume sufficient Oxidant is supplied. 08

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KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY

Department of Energy Science and Engineering

B. Sc. Engineering 4th Year 2nd Term Examination, 2020

ESE 4219

(Energy Audit and Management)

Time: 1 Hour 30 Minutes.

Full Marks: 120

- N.B. i) Answer any two questions from each section.
ii) Figures in the right margin indicate full marks.
iii) Assume reasonable data if any missing.

Section A

- 1(a). Define energy audit. What are the primary objectives of energy management? 08
- 1(b). Briefly describe the different elements that are included in an energy audit report. 12
- 1(c). For the following energy audit instruments, mention their working principle 10
(i). Combustion gas analyzer (ii). Ultrasound flow meter
- 2(a). What is normalozation of data? How does TPM maximize the productivity of manufacturing equipment? 10
- 2(b). In an industrial application, temperature is kept at 800°C. Requirements indicate that insulation material should not promote corrosion. Which insulation material would be used in this case? Also, mention three applications where your prescribed insulation material can be used. With mentioning the name of each curve draw the appropriate curves and mark the point of most economical insulation thickness. 10
- 2(c). How does green constrution reduces the use of energy and pure material? Can we use lead in green construction? Justify your answer. 10
- 3(a). Describe the responsibilities and duties of an energy manager. 10
- 3(b). Describe how ISO 5001 will work as a system to enhance energy efficiency. 10
- 3(c). Briefly discuss the following terms: 10
(i). Clean development mechanism (CDM)
(ii). Sustainable development

Section B

- 4(a). Explain briefly the operation of an ESCO. 08
- 4(b). Describe the following terms: 10
(i). Initial capital cost or net investment (ii). Cash flow diagram
- 4(c). What are the different criteria for financial an appraisal? Evaluate the financial merits of the two proposed projects shown in the following table using net present value analysis. The annual discount rate 8% for each project. 12

	Project I	Project II
Capital cost (Tk)	30,000	30,000
Year	Net annual saving(Tk)	Net annual saving(Tk)
1	+ 6,000	+ 6,600
2	+ 6,000	+ 6,600
3	+ 6,000	+ 6,300
4	+ 6,000	+ 6,300
5	+ 6,000	+ 6,000
6	+ 6,000	+ 6,000
Total net saving at the end of 6th year	+ 12,000	+ 12,000

- 5(a). What are the factors that must be taken under consideration during the technical design and financing of a project? 12
- 5(b). Which agreement was signed in June 1992 on climate change? Briefly explain the objectives of that agreement. 08
- 5(c). What is CUMSUM? What are the steps for analysis CUMSUM? 10
- 6(a). Why capacitors are added to the power distribution system? How a motor can be operated in an energy efficient way? 12
- 6(b). Which energy saving opportunity of lighting system is ignored in air conditioned space? How this system can be incorporated? 10
- 6(c). Suppose as an energy auditor, you are conducting an audit in a commercial building. Give a sample scope of your work for HVAC system of that building. 08

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KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY

Department of Energy Science and Engineering

B. Sc. Engineering 4th Year 2nd Term Examination, 2020

ESE 4013

(Energy in Built Environment)

Time: 1 Hour 30 Minutes

Full Marks: 120

N.B. i) Answer any TWO questions from each section.

ii) Figures in the right margin indicate full marks.

iii) Assume reasonable data if any missing.

SECTION – A

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|------|---|----|
| 1(a) | What is meant by Indoor Environment Control of a building? What are the internal and external factors upon which energy use of a building depend? | 10 |
| 1(b) | What are the four major aspects of indoor environment quality? How can ventilation contribute to maintain good indoor air quality? | 10 |
| 1(c) | What are the main sources of indoor air contamination? | 10 |
| 2(a) | What are meant by earth rotation and earth revolution? Explain with necessary figure. | 10 |
| 2(b) | How is built environmental influenced by the orientation of a building? Explain. | 12 |
| 2(c) | How can energy use for lighting be reduced? | 08 |
| 3(a) | Why is control of wind motion necessary for a building? Draw four wind flow patterns around a building which can help to increase air flow. | 06 |
| 3(b) | What are the functions of sun shading devices? Describe how to select a sun shading device for a building. | 05 |
| 3(c) | Classify the daylighting system. In contrast of daylighting, write short note on
i. Glazing specification
ii. Window height and location
iii. Tubular daylight devices | 16 |

SECTION – B

- | | | |
|------|--|----|
| 4(a) | Distinguish between 'low-e coating' and 'selective low-e'. | 05 |
| 4(b) | What is sol-air temperature? Explain the effect of absorptivity of the surface on sol-air temperature. | 08 |
| 4(c) | How does color affect heat transfer through walls? | 09 |
| 4(d) | Write a short note on "smart window". | 08 |
| | | |
| 5(a) | What are the major components of internal heat gains in a building? | 11 |
| 5(b) | Distinguish between infiltration and ventilation. How to prevent infiltration? | 12 |
| 5(c) | Why should building heat load be determined under transient condition? Explain. | 07 |
| | | |
| 6(a) | Explain the degree-day method to calculate the annual energy consumption of building. What are its advantages over bin method? | 10 |
| 6(b) | What is the basic difference between USA and European Union on the definition of zero-energy building? Why we need net-zero energy building? | 10 |
| 6(c) | What are the principles of net-zero energy building design? Explain. | 10 |

KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY

Department of Energy Science and Engineering

B. Sc. Engineering 4th Year 2nd Term Examination, 2020

ESE 4017

(Energy Efficiency Assessment)

Time: 1 hour 30 minutes

Full Marks: 120

- N.B. i) Answer any two questions from each section.
ii) Figures in the right margin indicate full marks.
iii) Assume reasonable data if any missing.

SECTION – A

- 1(a). Why indirect method of boiler efficiency calculation gives more accurate result than direct method? Explain. 05
- 1(b). Write down the advantages and disadvantages of (i) continuous blow down, and (ii) intermittent blow down. 08
- 1(c). The following are the data collected for a boiler using coal as the fuel. Find out the boiler efficiency by indirect method. 17

O_2 in flue gas = 9%

CO in flue gas = 0.08 %

CO_2 in flue gas = 11.1 %

Average flue gas temperature = 191°C

Ambient temperature = 31°C

Humidity in ambient air = 0.0198 kg/ kg of dry air

Mean specific heat of flue gas = 0.25 kcal/ kg.K

Mean specific heat of steam in flue gas = 0.44 kcal/ kg. K

Actual mass of air supplied = 12.3 kg/ kg of coal

Mass of dry flue gas = 12.9 kg/ kg of coal

Fuel analysis:

Carbon = 53.65 %

Hydrogen = 3.25%

Nitrogen = 1.30%

Oxygen = 8.57%

Sulphur = 0.34%

Moisture = 14.43%

Ash content = 18.54%

GCV of coal = 4100 kcal/ kg.

- 2(a). Why treatment of feed water is necessary for using in boiler? Briefly describe the “ Ion Exchange Process” of boiler water treatment. 12
- 2(b). What are the different approaches to save the energy used in boiler? Describe the “Recuperator” system to recover the waste heat from a boiler. 13
- 2(c). What is a steam distribution system? How to maintain right quantity and right pressure of end steam in any plant? 05
- 3(a). What are meant by fan curve and system characteristics curve? How to determine the operating point of a fan? Explain with necessary figures. 12
- 3(b). Classify fans, blowers and compressors according to ASME criteria. What are the differences between centrifugal and axial fans. 08
- 3(c). A fan is operating at 985 RPM, developing a flow of $23.35 \text{ m}^3/\text{s}$ at a static pressure of 515 mm WC . What will be the flow and static pressure if the speed is increased to 1350 RPM? 10

SECTION – B

- 4(a). Explain parallel and series operation of pumps with necessary pump curves. 08
- 4(b). A centrifugal pump pumping water operates at $35 \text{ m}^3/\text{hr}$ and at 1440 RPM. The pump operating efficiency is 68% and motor efficiency is 90%. The discharge pressure gauge shows 4.4 kg/cm^2 . The suction is 2.0 m below the pump centerline. If the speed of the pump is reduced by 50%, estimate the (i) new flow, (ii) head, and (iii) power. 10
- 4(c). Discuss the energy conservation opportunities in a pumping system. 12
- 5(a). If a pump runs 8000 hr/yr , describe how to calculate electricity expense from field measurements. 08
- 5(b). What are the sources of waste heat recovery in diesel generator system? Waste heat recovery from which source is more economically attractive and why? 10
- 5(c). Discuss energy saving measures for diesel generating sets. 12
- 6(a). Discuss in brief the different heat transfer loops in an industrial refrigeration system with schematic diagram. 10
- 6(b). Define Following terms: 8
- i) EER
- ii) IPLV
- 6(c). Write short description on components of a cooling tower. 12

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KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY

Department of Energy Science and Engineering

B. Sc. Engineering 4th Year 2nd Term Examination, 2020

ESE 4213

(Nuclear Power Engineering)

Time: 1 Hour 30 Minutes

Full Marks: 120

N.B. i) Answer any TWO questions from each section.

ii) Figures in the right margin indicate full marks.

iii) Assume reasonable data if any missing.

SECTION – A

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|------|---|----|
| 1(a) | Why is neutron used in nuclear reaction? Draw a fission chain reaction of Uranium-235. | 08 |
| 1(b) | "If binding energy per nucleon is relatively large, the nucleus is relatively stable." Justify the statement with necessary figure | 13 |
| 1(c) | Calculate the binding energy per nucleon of Helium-4. Note that the atomic mass of Helium-4 is $4.002602u$ | 09 |
| | | |
| 2(a) | Write short notes on α -decay and Gamma decay. Briefly describe why Gamma decay possess is more danger to us. | 10 |
| 2(b) | What is fundamental safety objective? Why safety of a nuclear power plant is so crucial, explain. | 10 |
| 2(c) | How does a government play role to attain safety of a nuclear power plant? Discuss the safety principle "Justification of facilities and activities" in your own words. | 10 |
| | | |
| 3(a) | With a schematic diagram, describe nuclear fuel cycle. | 15 |
| 3(b) | Discuss the advantages of in situ leach (ISL) mining process over open pit mining process. | 05 |
| 3(b) | Describe the waste conditioning method of sludge type radioactive waste. | 10 |

SECTION – B

- | | | |
|------|--|----|
| 4(a) | Classify nuclear reactor according to their evolution. Differentiate between PWR and BWR. | 11 |
| 4(b) | Write down the functions of moderator and control rod. | 08 |
| 4(c) | What is a passive safety system? Discuss the unique features of AP 1000 reactor design. | 11 |
| 5(a) | Write down the objectives of generation IV reactors. Describe the working principle of sodium-cooled fast reactor. | 10 |
| 5(b) | What is the loop of a nuclear power plant? Draw a flow diagram of the primary circuit of a four-loop plant. | 10 |
| 5(c) | What are the reactor vessel internals? Write short notes on lower core support structures. | 10 |
| 6(a) | How is a fuel rod formed? Describe with a neat sketch | 15 |
| 6(b) | How does the pressurizer limit pressure change during transients? Describe. | 15 |